

ALGORITHMS & DATA STRUCTURES (IT) REVISION LECTURE

2025–26

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Computing Science → Algorithms & Data Structures (IT)

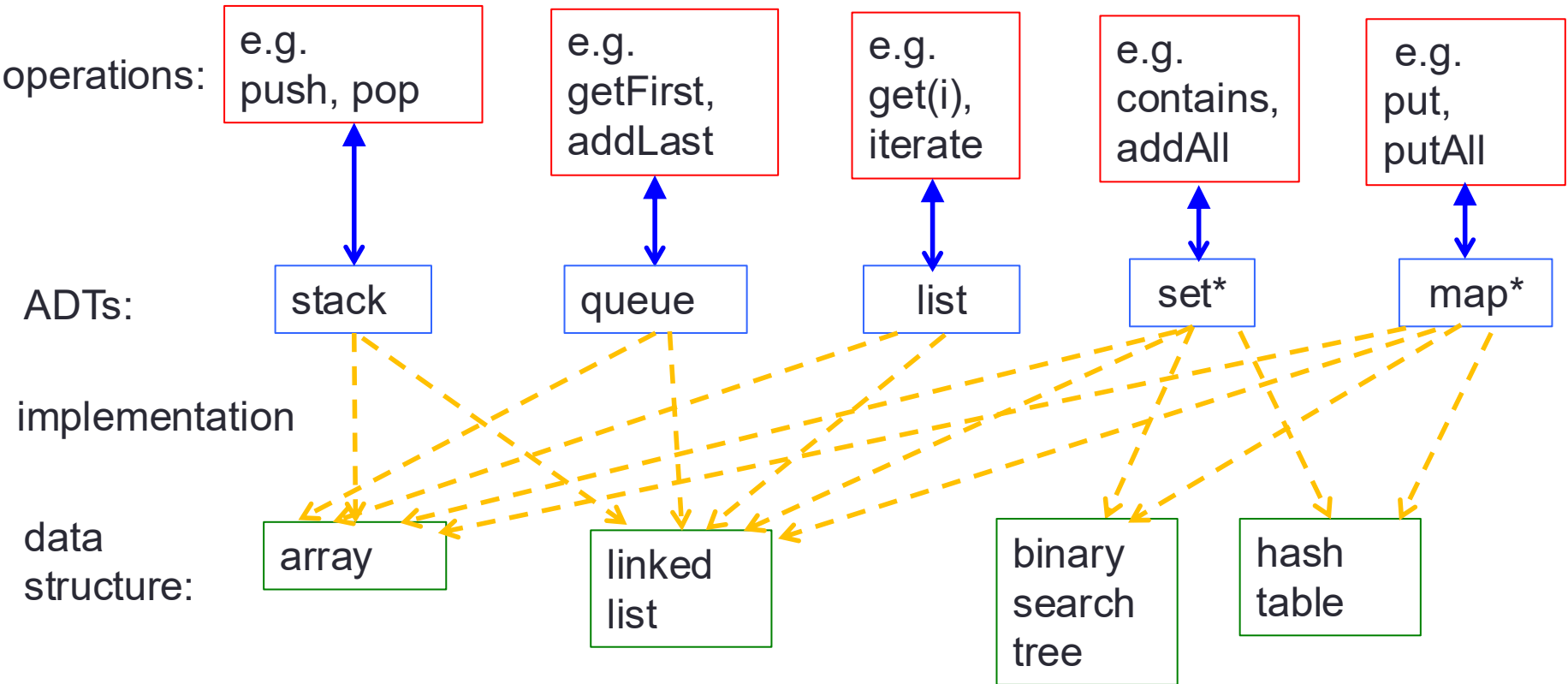
Aims – were these fulfilled?

- To study the concept of **algorithms**, and how to analyze their efficiency.
- To study the concept of **abstract data types**, and the abstract data types most commonly used in software development (stacks, queues, lists, sets, maps, trees, graphs).
- To study the basic **data structures** most commonly used to represent these abstract data types (arrays, linked-lists, binary-search-trees, hash-tables), together with algorithms operating on these data structures.

Contents (what we actually did)

1. Introduction
2. Algorithms and Complexity
3. The Array Data Structure
4. Linked-List Data Structures
5. Abstract Data Types (ADTs)
6. Stack ADTs
7. Queue ADTs
8. List and Iterator ADTs
9. Set ADTs
10. The Binary-Search-Tree Data Structure
- 10A. *AVL Trees*
11. Map ADTs
12. Hash-Table Data Structures
13. General algorithms

Final diagram



* + special boolean/value array implementations when entries/keys are small integers – equivalent to HTs when keys have this form

Format of exam: 1st May 14:00 in labs

3 questions, 60 marks in total

Answer all questions

Each of the 3 questions is composed of 10 parts. [All multiple choice](#)

Sample papers and past exam are given on the Moodle page. Exactly like the format of the actual exam (but different questions!)

Nb: Content and style of previous exams not all the same.

Format of exam contd:

It's in the form of a moodle quiz

- Not deliberately “out to get you”
- But you will need to think
- Take time to read the question and examine the material provided in boxes (see sample exam)
- You will have a handout sheet with some of the figures used in the exam to help you avoid excessive scrolling, and some rough paper.
- Negative marking 2 marks for a correct answer, -2/3 for an incorrect answer. 0 for no answer
- There are a lot of questions – don't be surprised if you can't finish them all.
- Options are jumbled
- No collaboration (will be severe penalty otherwise)

General advice

Don't leave it until the last minute to submit

- You may panic and submit incorrectly
- See instructions about good cause claim – you may have to take the resit exam in August if this happens

Don't try and look at others' papers

- Options are shuffled
- You will inherit negative marks from other people
- This exam is about *you*. It is not a joint effort
- *You will sign a form to say that you won't do this*

4. Time and space complexity

- Don't try and learn tables of complexities, but make sure you understand them.
- Obviously, you won't be asked to prove anything. But know what the recursive formulas lead to.
- You will always be told what n refers to (e.g. size of array, number of elements in a collection), and whether you are being asked for best, average or worst case complexity, and whether an array is sorted.
- The only time we have really thought about space complexity is for comparison of mergesort and quicksort.
- If asked about the complexity of an algorithm that looks like one you are familiar with, make sure you work your way through the algorithm in case it is a subtle variation.

4. How to revise

- Know the lecture slides inside out (i.e. where everything is, and understand them!)
- Go through the tutorial questions (not the implementation ones) and imagine the sort of multiple choice questions that could be related to them
- Go through the mock exam (solutions provided)
- Set multiple choice questions for each other!

RESIT EXAM

- Is not multiple choice
- 3 long questions
- Will be more like your other exams (I don't know if still in lab, presumably it will be).

Examinable material

- Everything apart from:
 - Expression Trees (in 10. BSTs), although useful in terms of seeing a good use of a stack
 - General algorithms – DFS, BFS algorithms (for graphs) in detail (you do need to know the bin packing material).